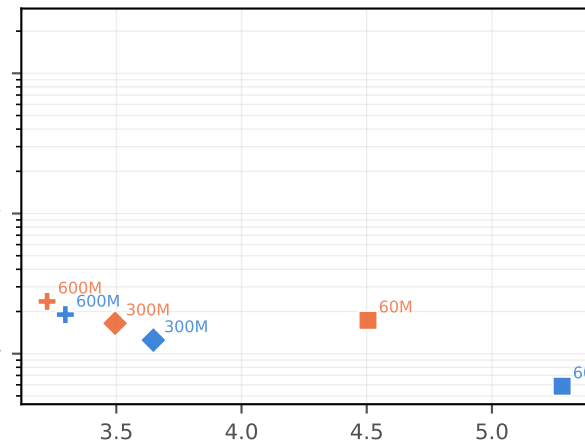


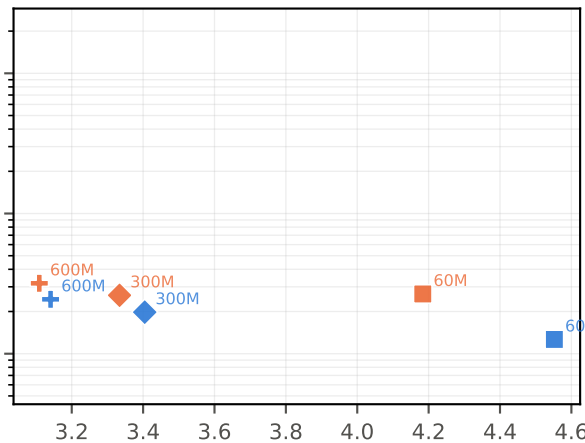
Relative degradation under Gaussian weight perturbation ($\gamma = 0.01$) vs model quality — one panel per chinchilla; small label = size

relative degradation (perturbed - base) / base

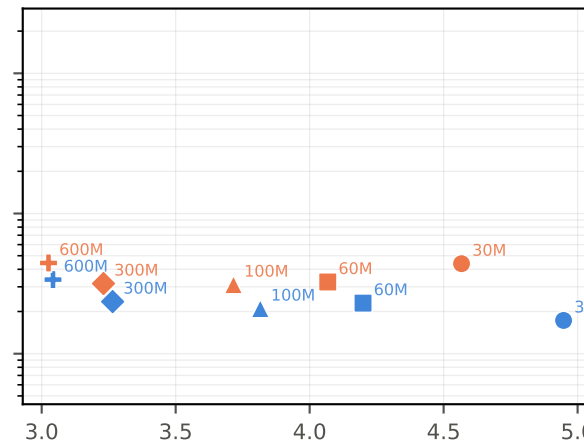
chinchilla 0.25



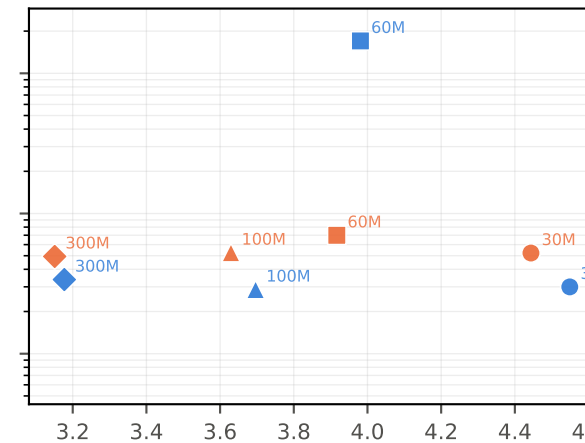
chinchilla 0.5



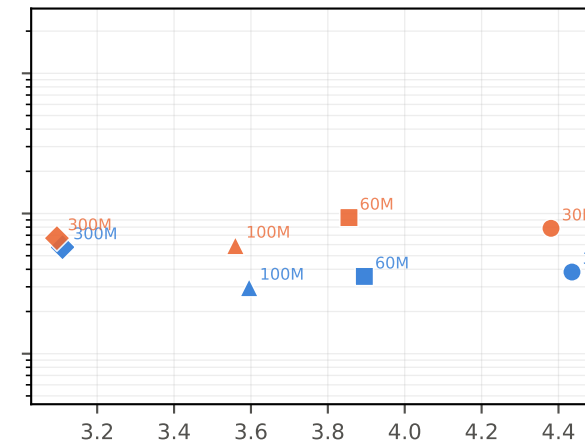
chinchilla 1



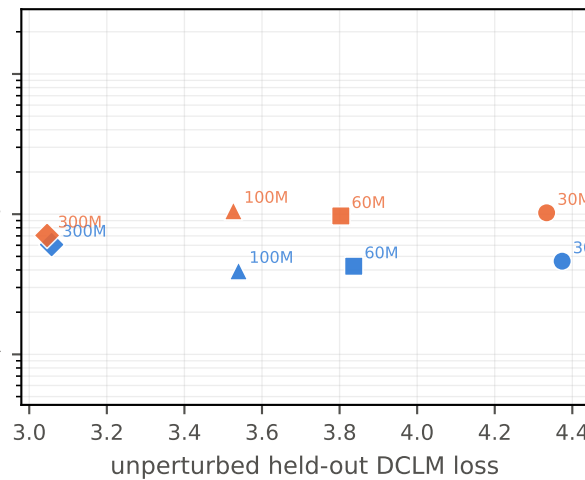
chinchilla 2



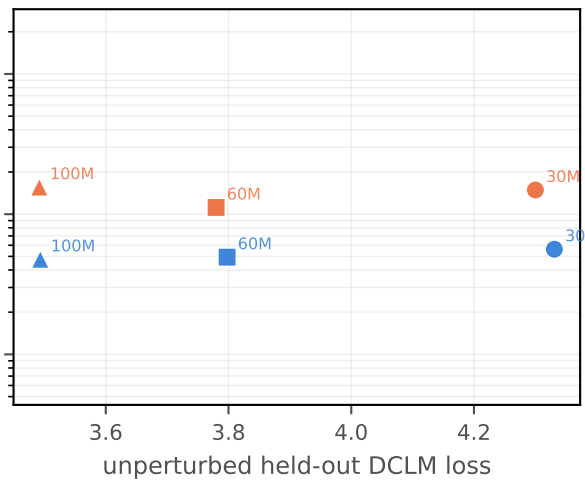
chinchilla 4



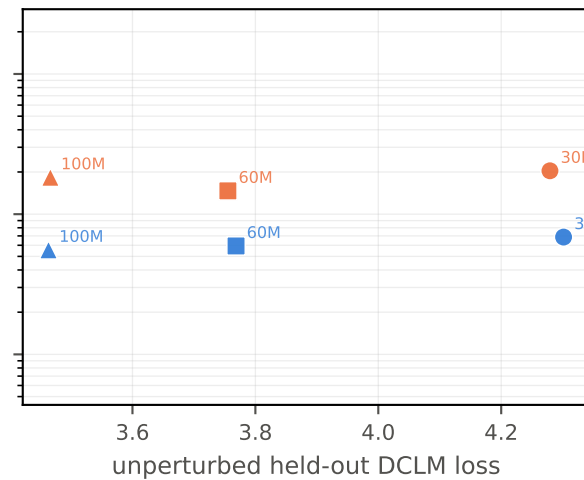
chinchilla 8



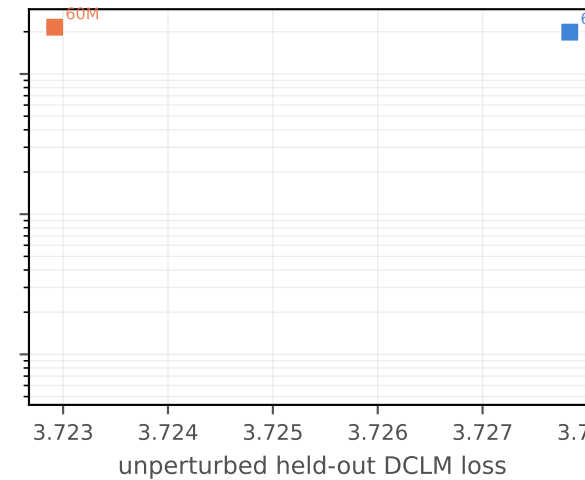
chinchilla 16



chinchilla 32



chinchilla 128



adamw muon 30M 60M 100M 300M 600M